JEE (ADVANCED) PHYSICS PRACTICE TEST

Topic: Mechanics | Time: 60 Mins | Total Questions: 2

Q1. [SINGLE_CHOICE] A motorboat going downstream overcame a raft at a point A. After time $\tau = 60\text{ min}$, the motorboat turned back and after some time passed the raft at a distance $l = 6.0\text{ km}$ downstream from point A. Assuming the duty of the engine to be constant and the river current uniform, find the river flow velocity v_r.

(A) A) 3.0 km/h

(B) B) 6.0 km/h

(C) C) 1.5 km/h

(D) D) 4.5 km/h

Q2. [SINGLE_CHOICE] A point moves in the xy-plane according to the equations $x = at$ and $y = at(1 - \alpha t)$, where a and α are positive constants, and t is time. Find the radius of curvature R of the trajectory as a function of x.

(A) A) $R = \frac{a}{2\alpha} \left[1 + \left(1 - \frac{2\alpha x}{a} \right)^2 \right]^{3/2}$

(B) B) $R = \frac{a}{\alpha} \left[1 + \left(1 - \frac{\alpha x}{a} \right)^2 \right]^{3/2}$

(C) C) $R = \frac{a}{2\alpha} \left[1 + \frac{4\alpha^2 x^2}{a^2} \right]^{3/2}$

(D) D) $R = \frac{2a}{\alpha} \left[1 + \left(1 - \frac{2\alpha x}{a} \right)^2 \right]^{1/2}$

ANSWER KEY & STEP-BY-STEP SOLUTIONS

Q1. Correct Answer: A

Solution: In the reference frame of the river water (which moves with velocity v_r relative to the bank):\n1. The raft is at rest.\n2. The motorboat moves away from the raft for time τ at speed $v_{b/r}$ (boat speed relative to water).\n3. When it turns back, it moves towards the raft with the same relative speed $v_{b/r}$.\n4. Therefore, the boat takes an equal time τ to return to the raft.\n5. Total time elapsed from point A until meeting the raft again is $T = 2\tau = 2\text{ hours}$.\n6. In this total time T, the raft has drifted a distance $l = 6.0\text{ km}$.\n7. Hence, $v_r = \frac{l}{T} = \frac{l}{2\tau} = \frac{6.0\text{ km}}{2\text{ h}} = 3.0\text{ km/h}$.

Q2. Correct Answer: A

Solution: 1. Express y in terms of x by eliminating time $t = \frac{x}{a}$:\n$$y = x \left(1 - \frac{\alpha x}{a}\right) = x - \frac{\alpha}{a}x^2$$\n2. Calculate the first derivative:\n$$\frac{dy}{dx} = 1 - \frac{2\alpha x}{a}$$\n3. Calculate the second derivative:\n$$\frac{d^2y}{dx^2} = -\frac{2\alpha}{a}$$\n4. Use the formula for the radius of curvature $R = \frac{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\left|\frac{d^2y}{dx^2}\right|}$:\n$$R = \frac{\left[1 + \left(1 - \frac{2\alpha x}{a} \right)^2 \right]^{3/2}}{\frac{2\alpha}{a}} = \frac{a}{2\alpha} \left[1 + \left(1 - \frac{2\alpha x}{a} \right)^2 \right]^{3/2}$$